

Design Notes – Single-Cell Li-Ion 3.3V / 500mA Regulator Board

1 Design Overview

This design implements a compact battery-powered 3.3V regulated power supply intended for embedded, RF, sensor, and low-noise analog applications. The board converts the output of a single-cell Li-ion battery (3.8V–4.2V nominal operating range) into a regulated 3.3V supply capable of delivering up to 500mA continuous output current.

The design incorporates reverse-polarity protection, transient suppression, overcurrent protection, power indication, thermal management, and dedicated test points for validation and debugging.

1.1 Key Specifications

Parameter	Value
Input Voltage	3.8V – 4.2V DC (Single-Cell Li-Ion)
Output Voltage	3.3V DC
Maximum Output Current	500mA
Typical Application Load	450mA
Operating Temperature	0°C to 70°C
PCB Layers	2
Form Factor	< 40mm × 20mm

2 IC Selection

2.1 LP5912-3.3DRVR (U1)

The LP5912-3.3DRVR low-noise LDO regulator from Texas Instruments was selected as the primary voltage regulation device.

Key advantages include:

- Fixed 3.3V output
- 500mA output current capability
- Very low dropout voltage
- Low output noise performance
- High power-supply rejection ratio (PSRR)
- Integrated thermal shutdown and current limiting
- Compact WSON package with exposed thermal pad

The LP5912 is particularly well suited for battery-powered applications where low noise and low dropout voltage are critical requirements.

3 Protection Circuitry

3.1 Reverse Polarity Protection (Q1)

Q1 (DMP2035U-7 P-Channel MOSFET) provides reverse-polarity protection at the battery input.

Compared to a conventional Schottky diode solution, the MOSFET introduces significantly lower voltage drop, preserving valuable headroom for the LDO regulator and maximizing battery utilization.

Benefits:

- Low forward voltage loss
- High efficiency
- No significant impact on dropout margin
- Automatic protection against reverse battery insertion

3.2 Overcurrent Protection (F1)

F1 (MF-NSMF110-2 Resettable PTC Fuse) provides input overcurrent protection.

The device protects the battery and PCB from excessive current during fault conditions such as:

- Output short circuits
- Component failures
- Input protection failures

The resettable nature of the fuse allows normal operation to resume automatically after fault removal.

3.3 Input Transient Protection (D1)

D1 (SMAJ5.0CA) is a bidirectional transient voltage suppressor (TVS) diode connected across the battery input.

Functions:

- ESD protection
- Hot-plug transient suppression
- Input surge suppression
- Protection against positive and negative voltage spikes

The device is located adjacent to the input connector to minimize transient propagation into the remainder of the circuit.

3.4 MOSFET Gate Protection (D2)

D2 (BZT52C6V8) is a 6.8V zener diode connected between the MOSFET gate and source.

Functions:

- Limits gate-to-source voltage
- Protects the MOSFET from gate overstress
- Improves robustness during transient events

4 Input and Output Capacitors

4.1 Input Capacitors (C1, C2)

C1 – 22 μ F Bulk Capacitor

Provides:

- Input energy storage

- Load transient support
- Battery impedance reduction

C2 – 100nF Ceramic Capacitor

Provides:

- High-frequency decoupling
- Noise suppression
- Improved regulator stability

Both capacitors are placed close to the LP5912 input pin to minimize parasitic inductance.

4.2 Output Capacitors (C3, C4)

C3 – 10 μ F Ceramic Capacitor

Primary output capacitor used for:

- Regulator stability
- Load transient response
- Output ripple reduction

C4 – 100nF Ceramic Capacitor

Provides:

- High-frequency output filtering
- Improved dynamic response

The capacitors are positioned directly adjacent to the regulator output and output connector.

5 Power Indicator

5.1 LED Indicator (D3)

A green LED is connected to the regulated 3.3V output rail through resistor R2.

Functions:

- Visual confirmation of regulator operation
- Output power status indication

The LED illuminates whenever the 3.3V rail is present.

6 Test Points

Dedicated test points are provided for validation and troubleshooting.

Test Point	Signal
TP1	VIN
TP2	VOUT
TP3	GND

These test points allow direct probing using multimeters and oscilloscopes without disturbing the circuit.

7 Thermal Design

The worst-case regulator dissipation occurs at maximum battery voltage and maximum load current.

Approximate worst-case dissipation:

$$P \approx (4.2\text{V} - 3.3\text{V}) \times 0.5\text{A} \approx 0.45\text{W}$$

The LP5912 exposed thermal pad is connected directly to the ground plane through a thermal-via array beneath the package.

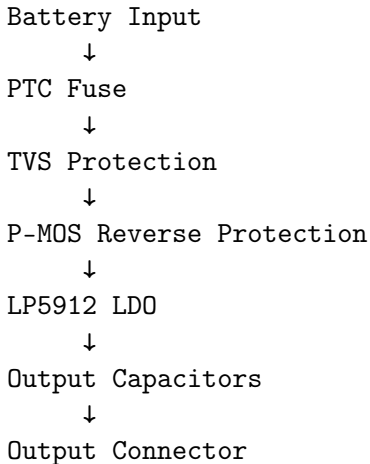
Thermal management features include:

- Exposed-pad grounding
- Thermal vias beneath U1
- Additional surrounding ground vias
- Continuous bottom-layer ground plane

These measures improve heat transfer into the PCB and maintain safe operating temperatures across the specified ambient temperature range.

8 PCB Layout Decisions

The PCB follows a left-to-right power flow architecture:



This arrangement minimizes power-path length and reduces voltage drop.

8.1 Layout Features

- 2-layer PCB construction
- Continuous ground plane on bottom layer
- Ground stitching vias around board perimeter
- Short power paths
- Wide copper regions for VIN and VOUT
- Thermal-via array beneath regulator
- Dedicated test points
- 45° routing practices
- Silkscreen labels for connectors and test points

The layout was optimized for manufacturability, thermal performance, and electrical robustness while remaining within the required board size constraints.

9 Conclusion

The completed design provides a compact and robust 3.3V power supply solution for single-cell Li-ion battery applications. The design integrates voltage regulation, reverse-polarity protection, transient suppression, overcurrent protection, thermal management, power indication, and debugging access while maintaining compatibility with standard low-cost 2-layer PCB fabrication processes. The design comfortably supports the specified 500mA output requirement and is suitable for powering embedded systems, RF modules, sensors, and low-noise analog circuitry.